IoT Temperature & Humidity Monitor

INTRODUCTION

The IoT Temperature and Humidity Monitoring System is a smart monitoring project developed using ESP8266 NodeMCU and DHT11/DHT22 sensor. This system measures temperature and humidity values from the environment and sends the data to the cloud using ThingSpeak platform.

OBJECTIVE

- To monitor temperature and humidity in real time
- To send sensor data to cloud platform
- To display live graphs using ThingSpeak
- To understand IoT communication and sensor interfacing

COMPONENTS REQUIREMENTS

ESP8266 NodeMCU

DHT11/DHT22 Sensor

Breadboard

Jumper Wires

10kΩ Resistor

USB Cable

Wi-Fi Connection

WORKING PRINCIPLE

The DHT11/DHT22 sensor reads environmental temperature and humidity values. The ESP8266 NodeMCU processes the sensor data and connects to Wi-Fi. The collected data is uploaded to ThingSpeak cloud platform every few seconds. Users can monitor live data and graphical analysis through the internet.

SOFTWARE USED

- Arduino IDE
- ThingSpeak Platform
- ESP8266 Board Package

APPLICATIONS

- Weather monitoring system
- Smart homes

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- Greenhouse monitoring
- Industrial monitoring
- Environmental analysis

CODE

```
#include <DHT.h>
#include <ESP8266WiFi.h>
#include <ThingSpeak.h>
#define DHTPIN D4 // GPIO2
#define DHTTYPE DHT22 // Change to DHT22 if using that sensor
const char* ssid = "V2022";
const char* password = "Sheetal2004";
const char* apiKey = "3LKQPBANHBWH2LEC";
unsigned long channelID = 3371801;
WiFiClient client;
DHT dht(DHTPIN, DHTTYPE);
void setup() {
Serial.begin(115200);
dht.begin();
WiFi.begin(ssid, password);
ThingSpeak.begin(client);
Serial.print("Connecting to Wi-Fi");
while (WiFi.status() != WL_CONNECTED) {
delay(500); Serial.print(".");
}
Serial.println(" Connected!");
}
void loop() {
float h = dht.readHumidity();
float t = dht.readTemperature();
```

```
if (isnan(h) || isnan(t)) {
Serial.println("Sensor read failed!"); return;
}
Serial.printf("Temp: %.1f C Humidity: %.1f %%\n", t, h);
ThingSpeak.setField(1, t);
ThingSpeak.setField(2, h);
int code = ThingSpeak.writeFields(channelID, apiKey);
Serial.println(code == 200 ? "Sent OK" : "Send failed: " + String(code));
delay(10000); // ThingSpeak free tier: min 15s; 10s is fine for testing
}
```